

Testing a new differentiator on two tasks: Dynamic Parameter Identification for Cable-Driven Parallel Robots and control of its motors

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Abstract : The aim of this paper is to study the practical interest of a new numerical differentiation method through two applications, the identification of numerical parameters of a robot and the control of its motors. The robot to be studied is a parallel cable-driven robot (CDPL). This paper first presents an initial work of identifying several dynamic parameters for cable-driven parallel robot (CDPR), *CRAFT* with a rigorous protocol. An orbital trajectory of the platform is designed in order to get a pure translation movement of the platform. This trajectory evolves in a plane and allows the identification of four dynamic parameters, the mass of the platform and the first three moments along the three main axes. From experimental data obtained with the moving-platform, the linear velocity and acceleration of the platform, which are necessary for parameters identification of the CDPR, are calculated with the application of an algorithm using semi-implicit homogeneous differentiators and compared with the application of the usual finite difference method. The obtained identification results show that in spite of the complexity of *CRAFT* an identification of all its essential dynamic parameters is possible with the two differentiation methods. Moreover, in the long term, an online identification of *CRAFT* during handling tasks is envisaged. In a second phase, the same two differentiation methods are used and compared to control in closed loop the eight motors such that the platform must track a orbital trajectory for three movement velocities.

1 Introduction

A CDPR consists of a moving-platform (MP) that is connected to a rigid frame by means of cables and actuators, the latter being generally mounted on the ground. These robots are very attractive for handling tasks [1] because of their low inertia, a higher payload to weight ratio and a large workspace compared to conventional manipulator robots with articulated rigid limbs. Their possible application fields can be industrial, or dedicated to search-and-rescue operations. To deal with various restrictions on cable tensions, cable elasticity, collisions and obstacle avoidance, over-actuation of the MP is actually a challenging scientific problem [2], [3].

The control of a CDPR is complex because, among other things, whatever its movement, the tension of its cables must always be positive. There are still open problems in the control of CDPRs. One key example of an open problem is that control design requires a consistent dynamic model with a relatively good knowledge of the dynamic parameters of the robot such as the terms of inertia, mass, and even friction at the actuator level, which manage the winding of the cables. Several contributions about identification of a robot cable exist. Kraus *et al.* [4] present an identification method for the complete actuator of a cable robot. A second-order system is established with a dead time as an analogous model. An inverse dynamic method (IDM) is used to identify the model parameters of a cable-driven finger joint for surgical robot [5]. But to our best knowledge the identification parameters of the handling platform of a cable robot during a movement in space is not currently investigated.

A CDPR, named *CRAFT* and located at LS2N, Centrale Nantes campus, is equipped with eight actuators and a MP. Each motor has an encoder sensor measuring the angular velocity of its output shaft allowing to evaluate the performances of the differentiation solutions. The MP has six degrees of freedom (DoF). This MP is thus over-actuated [6].

The identification of the dynamic parameters of a robot is essential to evaluate its behavior in simulation or to synthesize a control based model. The objective of the identification is not to find physical parameters of a mechanism as a robot, whose value is the most exact possible but to provide a modeling tool that is consistent in order to make simulation, control or other. The identification methodology is usually based on an inverse dynamic model that is linear as a function of the dynamic parameters to be identified [7], [8], or [9]. However the identification of dynamic parameters is not easy when the dynamic model is complex. Among the many challenges to be met is the search for exciting trajectories to identify the greatest number of dynamic parameters and overcome noise problems [10]. However the identification of dynamic parameters is not easy when the dynamic model is complex. The dynamic behavior of the *CRAFT* that moves in 3D space with rotation and translation combinations effectively requires a complex modeling with a lot of inertial parameters. In addition, one or more cables may slacken unexpectedly. As a consequence to initiate an identification work, only the model of the platform maneuvered by the cables is considered. Moreover, the defined trajectories generate only translation movements of the platform, *i.e.* without any rotational movement of the platform with respect to itself, in a plane. The model to describe the dynamic behavior of *CRAFT* becomes simpler with only four dynamic parameters to identify. To manage the noise problems semi-implicit Euler homogeneous differentiators are proposed (see e.g. [11] for a comparison of the techniques) and [12]. Off-line identification was conducted in order to compare, respectively from the planned trajectories of the platform, the measured filtered signals and their simplest Euler differentiation, and measured signals processed with an original semi-implicit homogeneous differentiators without extra filter. The main advantage of the semi-implicit Euler homogeneous differentiators is to offer the perspective of a real-time identification of dynamic parameters of robots and enhance the performance of closed-loop robot control. To do this, we need to consider the use of differentiators on line signals, and what is more, within a closed-loop control system for a mechanical system as complex as *CRAFT*. A preliminary evaluation of this differentiation method on real on-line signals is therefore necessary.

The contribution of this work is based on four parts :

1. Elaboration of a protocol in order to work with a dynamic model for translational movements of the platform and to have only four parameters to identify and experimental definition of this movement ;
2. Identification of the four parameters starting respectively from the reference trajectory, the cable tension measurement, the position measurement and the application of the simplest Euler differentiation to obtain the linear velocity and acceleration signals ;
3. These velocity and acceleration signals are compared with those obtained an original semi-

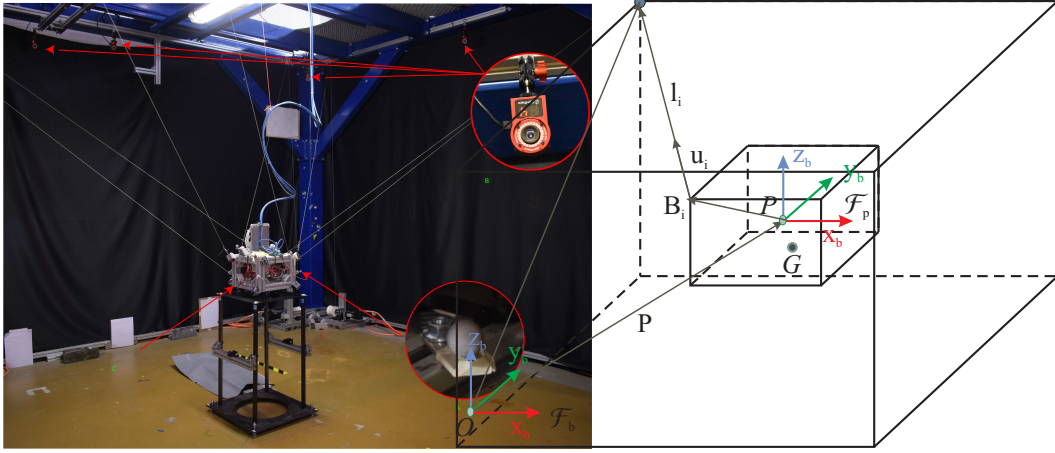


FIGURE 1 – (top) *bottom* prototype located at LS2N, Nantes, France and (right) *CRAFT* Geometric Parametrization.

implicit Euler homogeneous differentiator. The long-term idea of this work is to develop a complete framework that brings together numerical identification and derivation tools to identify the dynamic parameters of the on-line *CRAFT* as it performs handling tasks.

4. Furthermore in order to evaluate deeper its efficiency a comparison is made on-line between this original semi-implicit Euler homogeneous and the finite difference method thanks a feedback control law.

This paper is an extension of the work that was presented at the conference [13]. The long-term idea of this work is to develop a complete framework that brings together numerical identification and derivation tools to identify the dynamic parameters of the on-line *CRAFT* as it performs handling tasks and enhance the control of its motors.

The remaining of the paper is outlined as follows. Section 3 is devoted to the presentation of *CRAFT* namely, its geometric, kinematic, dynamic model and its adaptation to the presented strategy. A semi-implicit homogeneous differentiator, which is considered in this study to get the velocities and the accelerations from the measured position, is presented Sec. 4. The identification methodology, which is based on the least-squares (*LS*) estimator is presented in Sec. 5, with the solving way. The feedback control to test the original semi-implicit Euler homogeneous differentiator is defined in Sec. 6. The experimental results are presented in Sec. 7 for both tasks, identification and control, with a comparison between the usual finite difference method and the homogeneous semi-implicit differentiator. Conclusions and future work are offered in Sec. 8.

2 Robot *CRAFT*

The cable-driven parallel robot prototype, named *CRAFT* is located at LS2N, France. The base frame of *CRAFT* is 4 m long, 3.5 m wide, and 2.7 m high, see Figure 1. The three- translational and the three- rotational motions of its suspended MP are controlled with eight cables being respectively wound around eight actuated reels fixed to the ground. The MP is 0.28 m long, 0.28 m wide, and 0.2 m high, its overall mass being approximately equal to 10 kg. For each of the eight electrical motors an encoder sensor with 2000 pulses per revolution, measures the angular variable of its shaft. The eight motors are equipped with a gearbox reducer of ratio $n_r = 8$. The measured value with the encoder sensor is divided by n_r in order to obtain the angular position of the output shaft of the gearbox reducer. The robot *CRAFT* has no tachometer.

The main hardware of the prototype consists of a PC (equipped with © MATLAB and © Control-Desk software), eight © PARKER SME60 motors and TPD-M drivers, a © dSPACE DS1007-based real-time controller and eight custom made winches. Each cable can exert a tension up to 150 N to the MP. The maximum velocity of each cable is equal to 5.9 m/s. The cable tensions are measured using eight *FUTEK FSH04097* sensors, one for each cable, attached to cable anchor points. Their signal is

amplified using eight FSH03863 voltage amplifiers and sent to the robot controller by a coaxial cable. *CRAFT* is equipped with a motion capture, *OptiTrack* that the sampling period is equal to 0.01 s.

3 Robot Modeling

Figure 1 depicts the main geometric parameters of *CRAFT* and the i^{th} cable where $i \in \{1, \dots, m\}$, m being the number of cables attached to the n DoF MP (for *CRAFT*, $m = 8$ and $n = 6$). The dynamic equilibrium wrench equation can be expressed as :

$$\mathbf{W}\boldsymbol{\tau} - \mathbb{I}_p\ddot{\mathbf{p}} - \mathbf{C}\dot{\mathbf{p}} + \mathbf{w}_g = \mathbf{0}_m \quad (1)$$

with $\mathbf{W}$ being the robot wrench matrix expressed as :

$$\mathbf{W} = \begin{bmatrix} \mathbf{u}_1 & \dots & \mathbf{u}_i & \dots & \mathbf{u}_m \\ \mathbf{b}_1 \times \mathbf{u}_1 & \dots & \mathbf{b}_i \times \mathbf{u}_i & \dots & \mathbf{b}_m \times \mathbf{u}_m \end{bmatrix} \quad (2)$$

where $\mathbf{u}_i$ is the unit vector of the i^{th} cable vector pointing from cable anchor points B_i to exit points A_i , expressed in the reference frame $\mathcal{F}_b$. Vector $\mathbf{b}_i$ pointing from point P to point B_i and is expressed in the platform frame $\mathcal{F}_p$. From Eq. (1), $\boldsymbol{\tau} = [\tau_1, \dots, \tau_i, \dots, \tau_m]$ is the cable tension vector, $\mathbb{I}_p$ is the inertia tensor, $\mathbf{C}$ is the Coriolis matrix and $\mathbf{w}_g$ is the gravity wrench. $\dot{\mathbf{p}} = [\dot{\mathbf{t}}, \boldsymbol{\omega}]^\top$ and $\ddot{\mathbf{p}} = [\ddot{\mathbf{t}}, \boldsymbol{\alpha}]^\top$ are the vectors of the MP velocity and acceleration, where, $\dot{\mathbf{t}} = [\dot{t}_x, \dot{t}_y, \dot{t}_z]^\top$ and $\ddot{\mathbf{t}} = [\ddot{t}_x, \ddot{t}_y, \ddot{t}_z]^\top$ are the vectors of the MP linear velocity and acceleration, while $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^\top$ and $\boldsymbol{\alpha} = [\alpha_x, \alpha_y, \alpha_z]^\top$ are the vectors of the MP angular velocity and angular acceleration, respectively. The gravity wrench is :

$$\mathbf{w}_g = M \begin{bmatrix} \mathbf{I}_3 \\ {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \end{bmatrix} \mathbf{g} \quad (3)$$

where M is the MP mass, $\mathbf{I}_3$ is the 3×3 identity matrix, ${}^b\mathbf{R}_p$ is the rotation matrix of the MP $\mathcal{F}_p$ frame w.r.t the base frame $\mathcal{F}_b$, $\hat{\mathbf{S}}_G$ is the skew-symmetric matrix associated to $\mathbf{s}_G = [x_G, y_G, z_G]^\top$ which represents the coordinate vector of the center of mass of the MP expressed in $\mathcal{F}_p$ as :

$$\hat{\mathbf{S}}_G = \begin{bmatrix} 0 & -z_G & y_G \\ z_G & 0 & -x_G \\ -y_G & x_G & 0 \end{bmatrix} \quad (4)$$

$M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G$ represents the first momentum of the MP defined with respect to base frame $\mathcal{F}_b$. $\mathbf{g} = [0, 0, -g]$ is the gravity acceleration vector where $g = 9.81 \text{ m s}^{-2}$. The matrix $\mathbb{I}_p$ in Eq. (1) can be expressed as :

$$\mathbb{I}_p = \begin{bmatrix} M \mathbf{I}_3 & -M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \\ M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G & \mathbf{I}_p \end{bmatrix} \quad (5)$$

with $\mathbf{I}_p$ being the MP inertia tensor expressed in $\mathcal{F}_p$. The term $\mathbf{C}$, which represents the centrifugal and Coriolis effects in Eq. (1) is expressed as :

$$\mathbf{C} = \begin{bmatrix} \mathbf{0}_3 & -M \hat{\boldsymbol{\omega}} \hat{\mathbf{S}}_G \\ \mathbf{0}_3 & \hat{\boldsymbol{\omega}} \mathbf{I}_p \end{bmatrix} \quad (6)$$

Using Eqs. (3), (4), and (6) in Eq. (1), gives :

$$\begin{bmatrix} M \mathbf{I}_3 & -M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \\ M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G & \mathbf{I}_p \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{t}} \\ \boldsymbol{\alpha} \end{bmatrix} + \begin{bmatrix} \mathbf{0}_3 & -M \hat{\boldsymbol{\omega}} \hat{\mathbf{S}}_G \\ \mathbf{0}_3 & \hat{\boldsymbol{\omega}} \mathbf{I}_p \end{bmatrix} \begin{bmatrix} \dot{\mathbf{t}} \\ \boldsymbol{\omega} \end{bmatrix} - M \begin{bmatrix} \mathbf{I}_3 \\ {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \end{bmatrix} \mathbf{g} = \mathbf{W}\boldsymbol{\tau} \quad (7)$$

As a first step of the identification, in this article, we will only consider platform translations, *i.e.*, the platform does not rotate in relation to itself. Hence, the angular velocities and accelerations of the MP in Eq. (7) vanish and is reduced to :

$$\begin{bmatrix} M \mathbf{I}_3 \\ M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \end{bmatrix} \ddot{\mathbf{t}} - \begin{bmatrix} M \mathbf{I}_3 \\ M {}^b\mathbf{R}_p \hat{\mathbf{S}}_G \end{bmatrix} \mathbf{g} = \mathbf{W}\boldsymbol{\tau} \quad (8)$$

As the platform orientation is always constant, we can consider ${}^b\mathbf{R}_p = \mathbf{I}_3$.

$${}^b\mathbf{R}_p = \begin{bmatrix} r_1 & r_2 & r_3 \\ r_4 & r_5 & r_6 \\ r_7 & r_8 & r_9 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (9)$$

Eq. (8) can be expressed as in the form as :

$$\begin{bmatrix} M\ddot{t}_x \\ M\ddot{t}_y \\ M\ddot{t}_z \\ -M\ddot{t}_y z_G + M\ddot{t}_z y_G \\ M\ddot{t}_x z_G - M\ddot{t}_z x_G \\ -M\ddot{t}_x y_G + M\ddot{t}_y x_G \end{bmatrix} - \begin{bmatrix} 0 \\ 0 \\ -Mg \\ -My_G g \\ Mx_G g \\ 0 \end{bmatrix} = \mathbf{W}\boldsymbol{\tau} \quad (10)$$

$$\begin{bmatrix} M\ddot{t}_x \\ M\ddot{t}_y \\ M\ddot{t}_z + Mg \\ -M\ddot{t}_y z_G + M\ddot{t}_z y_G + My_G g \\ M\ddot{t}_x z_G - M\ddot{t}_z x_G - Mx_G g \\ M\ddot{t}_y x_G - M\ddot{t}_x y_G \end{bmatrix} = \mathbf{W}\boldsymbol{\tau} \quad (11)$$

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad (12)$$

where,

$$\mathbf{A} = \begin{bmatrix} \ddot{t}_x & 0 & 0 & 0 \\ \ddot{t}_y & 0 & 0 & 0 \\ \ddot{t}_z + g & 0 & 0 & 0 \\ 0 & 0 & \ddot{t}_z + g & -\ddot{t}_y \\ 0 & -\ddot{t}_z - g & 0 & \ddot{t}_x \\ 0 & \ddot{t}_y & -\ddot{t}_x & 0 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} M \\ Mx_G \\ My_G \\ Mz_G \end{bmatrix}, \quad \mathbf{b} = \mathbf{W}\boldsymbol{\tau} \quad (13)$$

Here $[Mx_G \ My_G \ Mz_G]^\top$ is the first moment of inertia of MP about the origin of the MP frame $\mathcal{F}_p$. In Eq. (13), we need the linear acceleration of the MP $\ddot{\mathbf{t}}$ and the cable tensions $\boldsymbol{\tau}$. As the robot is already equipped with the cable tension sensors so we can directly acquire $\boldsymbol{\tau}$.

4 Semi-implicit Homogeneous differentiator

The purpose is to estimate the velocity of the angular variable exclusively from the measured position of the output shaft for each of the eight motors using a discretized semi-implicit Homogeneous differentiator. To present this differentiator let us introduce the continuous-time state :

$$\Sigma_1 : \begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= p(t) \\ y &= x_1 \end{cases} \quad (14)$$

where x_1 and x_2 are respectively the angular variable and its angular velocity. Here $p(t)$ is a bounded perturbation as it satisfies :

$$\exists p_M > 0 \text{ such that } |p(t)| < p_M \text{ for all } t > 0 \quad (15)$$

y is the measure signal of $x_1, y(t) \in \mathcal{C}^\omega$ i.e. assumed as an analytic signal :

$$y(t+h) = y(t) + \sum_{j=1}^{\infty} y^{(j)}(t) \frac{h^j}{j!} \quad (16)$$

The variable p is considered slowly variable with respect to the sampling period h i.e. $\forall t > 0$ and $\forall \delta \in [0, h]$

$$p(t + \delta) - p(t) \simeq 0$$

Then, for time periods of the order of h , equation (14) can be rewritten as follows in term of triple integrator without input :

$$\Sigma_2 : \begin{cases} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ \dot{x}_3 &= 0 \\ y &= x_1 \end{cases} \quad (17)$$

where $x_3 := p$.

In order to design a differentiator Homogeneity approach is interesting because if a local stability is obtained due to the dilatation, this framework allows extending this local property to global settings, [14]. This differentiator can be written as [15], [16], [17],

$$\begin{cases} \dot{z}_1 = z_2 + \lambda_1 \mu \lceil \epsilon_1 \rceil^\alpha \\ \dot{z}_2 = z_3 + \lambda_2 \mu^2 \lceil \epsilon_1 \rceil^{2\alpha-1} \\ \dot{z}_3 = \lambda_3 \mu^3 \lceil \epsilon_1 \rceil^{3\alpha-2} \\ \hat{y} = z_1 \end{cases} \quad (18)$$

where $\epsilon_1 = y - z_1$; the notation $\lceil \bullet \rceil^\alpha = |\bullet|^\alpha \text{sgn}(\bullet)$ is adopted along the paper; $\alpha \in]0.5, 1[$ has to be fixed [18]. When $\alpha = 1$ the differentiator is linear, $\alpha = 0.5$ defines the proposed differentiator by Levant [19]. This differentiator has the best accuracy with respect to perturbation but it is more sensitive to noise. The choice $\mu^2 > p_M$ is made in order to cancel the effect of the unknown perturbation $p(t)$. The homogeneity degree d of the differentiator Eq. (18) is equal to $\alpha - 1$ with respect to dilatation Λ_r with $r = (r_1 = 1, r_2 = 1)$ [16]. Moreover, $\lambda_i > 0$, $i = 1, 2, 3$ are the linear part gains, and allow to have the eigenvalues of the differentiation error ϵ_1 in the left part of the complex plane i.e. the eigenvalues have a negative real part.

The unknown perturbation $p(t)$ being assumed to be a constant parameter or a slowly variable, this implies that for a sufficient small sampling-time $h > 0$, $p \equiv p^+$, with the notation for a discretized variable :

$$\begin{aligned} \bullet(t = (k+1)h) &= \bullet^+ \\ \bullet(t = kh) &= \bullet \end{aligned} \quad (19)$$

The implicit Euler discretization of the continuous-time model Eq. (14) can therefore be written with Eq. (19)

$$\begin{cases} x_1^+ = x_1 + h x_2^+ \\ x_2^+ = x_2 + h p^+ \\ x_3^+ = x_3 \\ y = x_1 \end{cases} \quad (20)$$

The discretization of the differentiator Eq. (18) is based on the so-called implicit projection that acts as corrective terms and aim to "generalize" the sign function in sliding-based differentiator in order to reduce the chattering and preserve stability properties for high time steps. Three projectors $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}_3$ are used respectively to design the correction terms with the differentiator SIHD-3. The projectors $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}_3$ are respectively dedicated to the estimation of z_1 , z_2 , and z_3 . Considering the signal to differentiate y , the error $\epsilon_1 = y - z_1$, and the definition of the notation $\lceil \epsilon_1 \rceil \equiv |\epsilon_1| \text{sign}(\epsilon_1)$, An iterative projector, denoted $\mathcal{N}_q$, where q is the index of differentiation, controls the estimation of

each state z_1 , z_2 , and z_3 . The semi-implicit homogeneous Euler discretization based on three projectors (SIHD-3) reads as :

$$\begin{cases} z_1^+ = z_1 + h(z_2^+ + \lambda_1 \mu |\epsilon_1|^\alpha \mathcal{N}_1) \\ z_2^+ = z_2 + E_1 h(z_3^+ + \lambda_2 \mu^2 |\epsilon_1|^{2\alpha-1} \mathcal{N}_2) \\ z_3^+ = z_3 + E_1 E_2 h \lambda_3 \mu^3 |\epsilon_1|^{3\alpha-2} \mathcal{N}_3 \end{cases} \quad (21)$$

where each line z_q^+ is associated respectively to the estimation of the position (index $q = 1$), the estimation of the velocity (index $q = 2$) and the estimation of the acceleration ($q = 3$). The associated projectors $\mathcal{N}_q$ with $q \in \{1, 2, 3\}$ are defined by :

$$\mathcal{N}_q := \begin{cases} \text{if } |\epsilon_1|^{q(1-\alpha)} < \lambda_q h^q \rightarrow \mathcal{N}_q = \frac{[\epsilon_1]^{q(1-\alpha)}}{\lambda_q h^q} \\ \text{if } |\epsilon_1|^{q(1-\alpha)} \geq \lambda_q h^q \rightarrow \mathcal{N}_q = \text{sgn}(\epsilon_1) \end{cases} \quad (22)$$

Moreover, E_q is defined for $q = \{1, 2\}$ such as :

$$E_q := \begin{cases} \text{if } |\epsilon_1|^{q(1-\alpha)} < \lambda_q h^q \rightarrow E_q = 1 \\ \text{if } |\epsilon_1|^{q(1-\alpha)} \geq \lambda_q h^q \rightarrow E_q = 0 \end{cases} \quad (23)$$

The variable E_q allows activating the corresponding $(q + 1)$ th line when the convergence domain has been reached at the q th line. This avoids unexpected behavior between the lines, such as oscillations that could destabilize the differentiator.

The filtering properties of the differentiator avoid using an extra filter before the differentiation. The measured angular positions are noisy such as y becomes $y_m = x_1 + \eta$. The λ_i , $i = 1, 2, 3$ parameters are chosen such as the linear part is stable. The value of homogeneous exponent α is chosen to allow better filtering properties of the estimated differentiation. The μ parameter is also chosen by numerical test trial and error to be equal to 1, in order to determine the best possible action of the three projectors $\mathcal{N}_1$, $\mathcal{N}_2$, and $\mathcal{N}_3$.

5 Identification Methodology

The identification methodology is based on the used of the inverse dynamic model (*ID*) Eq. (12), [7], [20], and [9]. This model is linear as function of the four parameters M , Mx_G , My_G , and Mz_G . An off-line identification of these parameters is then considered, given measured or estimated off-line data for the linear acceleration $\ddot{\mathbf{t}}$ of the MP and the cable tensions $\boldsymbol{\tau}$, collected while the robot *CRAFT* is tracking a planned trajectory. The model Eq. (12) is sampled at a sufficient number of time samples t_i , for $i = 1, \dots, n_e$, with $(6n_e) \gg 4$, in order to get an over-determined linear system of $(6n_e)$ equations [9] :

$$\mathcal{Y}_{ID}(\boldsymbol{\tau}) = \mathcal{W}_{ID}\mathbf{x} + \boldsymbol{\rho} \quad (24)$$

where

$$\mathcal{Y}_{ID}(\boldsymbol{\tau}) = \begin{pmatrix} b_1(6 \times 1) \\ \vdots \\ b_{n_e}(6 \times 1) \end{pmatrix} ; \quad \mathcal{W}_{ID} = \begin{pmatrix} A_1(6 \times 4) \\ \vdots \\ A_{n_e}(6 \times 4) \end{pmatrix}, \quad (25)$$

$\boldsymbol{\rho}$ is the $(6n_e \times 1)$ vector of errors between the data of the experimental measurement of the torques in $\mathcal{Y}_{ID}(6n_e \times 1)$ and the data $\mathcal{W}_{ID}(6n_e \times 4)$ predicted by the model. These errors are due to noise measurement and modeling error. The identification problem consists in finding $\mathbf{x}$ in order to minimize the norm squared of the error $\boldsymbol{\rho}$:

$$\|\boldsymbol{\rho}\|^2 = \|\mathcal{Y}_{ID} - \mathcal{W}_{ID}\mathbf{x}\|^2 \quad (26)$$

The *LS* estimator models a process by fitting the parameter vector $\mathbf{x}$ according to the minimization of Eq. (26). Then model (24) becomes

$$\mathcal{Y}_{ID}(\boldsymbol{\tau}) = \mathcal{W}_{ID}\hat{\mathbf{x}} + \hat{\boldsymbol{\rho}} \quad (27)$$

The estimated parameter $\hat{\mathbf{x}}$ is equal to :

$$\hat{\mathbf{x}} = \mathcal{W}_{ID}^+ \mathcal{Y}_{ID} \quad (28)$$

where $\mathcal{W}_{ID}^+$ is the pseudo-inverse of $\mathcal{W}_{ID}$. Since $6n_e > 4$ then

$$\mathcal{W}_{ID}^+ = (\mathcal{W}_{ID}^\top \mathcal{W}_{ID})^{-1} \mathcal{W}_{ID}^\top.$$

$\hat{\mathbf{x}}$ is the unique LS solution of Eq. (28). The standard deviation $\sigma_{\hat{\mathbf{x}}}$ is estimated assuming that $\mathcal{W}_{ID}$ is a deterministic matrix and ρ is a zero-mean additive independent Gaussian noise, with a covariance matrix $C_{\rho\rho}$

$$C_{\rho\rho} = E(\rho\rho^\top) = \sigma_\rho^2 \mathbf{I} \quad (29)$$

E is the expectation operator and $\mathbf{I}(6n_e \times 6n_e)$ is the identity matrix. By replacing $\mathcal{Y}_{ID}$ in Eq. (28) with its definition Eq. (24) we obtain :

$$\hat{\mathbf{x}} = \mathbf{x} + \mathcal{W}_{ID}^+ \rho \quad (30)$$

Applying the expectation operator at both sides of Eq. (30) ρ being a zero-mean additive independent Gaussian noise we get :

$$E(\hat{\mathbf{x}}) = E(\mathbf{x}) \quad (31)$$

Thus the estimated values are unbiased. An unbiased estimation of the standard deviation σ_ρ is :

$$\hat{\sigma}_\rho^2 = \frac{\|\mathcal{Y}_{ID} - \mathcal{W}_{ID} \hat{\mathbf{x}}\|^2}{(6n_e - 4)} \quad (32)$$

$\mathcal{W}_{ID}$ and ρ , being respectively a non-stochastic matrix and a stochastic matrix, the covariance matrix of the LS estimation error is then given by :

$$\begin{aligned} C_{\hat{\mathbf{x}}\hat{\mathbf{x}}} &= E((\mathbf{x} - \hat{\mathbf{x}})(\mathbf{x} - \hat{\mathbf{x}})^\top) \\ &= (\mathcal{W}_{ID}^\top \mathcal{W}_{ID})^{-1} \mathcal{W}_{ID}^\top \sigma_\rho^2 \mathbf{I} \mathcal{W}_{ID} (\mathcal{W}_{ID}^\top \mathcal{W}_{ID})^{-1} \\ &= \hat{\sigma}_\rho^2 (\mathcal{W}_{ID}^\top \mathcal{W}_{ID})^{-1} \end{aligned} \quad (33)$$

$\hat{\sigma}_{\mathbf{x}_i}^2 = C_{\hat{\mathbf{x}}\hat{\mathbf{x}}}(i, i)$ is the i^{th} diagonal coefficient of $C_{\hat{\mathbf{x}}\hat{\mathbf{x}}}$. The relative standard deviation $\% \hat{\sigma}_{\mathbf{x}_i}$ for each identified parameters is given by :

$$\% \hat{\sigma}_{\mathbf{x}_i} = \frac{\hat{\sigma}_{\mathbf{x}_i}}{|\hat{x}_i|} \quad (34)$$

The calculation of Eq. (24) and the condition number of $\mathcal{W}_{ID}$, $\text{Cond}(\mathcal{W}_{ID})$, can be obtained using the singular value decomposition (SVD) of $\mathcal{W}_{ID}$. [21].

$$\mathcal{W}_{ID} = \mathbf{U} \Sigma \mathbf{V}^\top \quad (35)$$

where $\mathbf{U}_{(6n_e \times 6n_e)} = \begin{pmatrix} U_{1(6n_e \times r)} & U_{2(6n_e \times (6n_e - r))} \end{pmatrix}$ and $\mathbf{V}_{(4 \times 4)} = \begin{pmatrix} V_{1(r \times r)} & V_{2(r \times (4-r))} \end{pmatrix}$ are orthogonal matrices. The matrix $\mathcal{W}_{ID}(6n_e \times 4)$ can be rewritten as follows :

$$\mathcal{W}_{ID} = \begin{pmatrix} U_1 & U_2 \end{pmatrix} \begin{pmatrix} \mathbf{S}_{r \times r} & \mathbf{0}_{r \times (4-r)} \\ \mathbf{0}_{(6n_e - r) \times r} & \mathbf{0}_{(6n_e - r) \times (4-r)} \end{pmatrix} \begin{pmatrix} V_1^\top \\ V_2^\top \end{pmatrix} \quad (36)$$

$\mathbf{S}$ is a diagonal matrix, formed by the singular values of $\mathcal{W}_{ID}$ *i.e.* the eigenvalues of $\mathcal{W}_{ID}^\top \mathcal{W}_{ID}$. The singular values are strictly positive. By multiplying Eq. (24) by $\mathbf{U}^\top$ gives :

$$\begin{aligned} \begin{pmatrix} U_1^\top \mathcal{Y}_{ID} \\ U_2^\top \mathcal{Y}_{ID} \end{pmatrix} &= \begin{pmatrix} \mathbf{S}_{r \times r} & \mathbf{0}_{r \times (4-r)} \\ \mathbf{0}_{(6n_e - r) \times r} & \mathbf{0}_{(6n_e - r) \times (4-r)} \end{pmatrix} \begin{pmatrix} V_1^\top \mathbf{x} \\ V_2^\top \mathbf{x} \end{pmatrix} \\ &\quad + \mathbf{U}^\top \rho \end{aligned} \quad (37)$$

The multiplication of a vector by an orthogonal matrix preserves the norm of this vector. The identification problem Eq. (26) is then equivalent to :

$$\begin{aligned}
\|\rho\|^2 &= \|\mathbf{U}^\top \mathcal{Y}_{ID} - \mathbf{U}^\top \mathcal{W}_{ID} \mathbf{x}\|^2 \\
&= \left\| \begin{pmatrix} U_1^\top \mathcal{Y}_{ID} \\ U_2^\top \mathcal{Y}_{ID} \end{pmatrix} - \begin{pmatrix} \mathbf{S}_{r \times r} V_1^\top \mathbf{x} \\ \mathbf{0}_{(6n_e - r) \times 4} \end{pmatrix} \right\|^2 \\
&= \|U_1^\top \mathcal{Y}_{ID} - \mathbf{S} V_1^\top \mathbf{x}\|^2 + \|U_2^\top \mathcal{Y}_{ID}\|^2
\end{aligned} \tag{38}$$

The minimal norm of ρ , $\|\rho\|_{\min} = \|U_2^\top \mathcal{Y}_{ID}\|$, is given for any value $V_2^\top \mathbf{x}$ when $U_1^\top \mathcal{Y}_{ID} - \mathbf{S} V_1^\top \mathbf{x}$ is canceled by the following unique solution :

$$\widehat{V_1^\top \mathbf{x}} = \mathbf{S}^{-1} U_1^\top \mathcal{Y}_{ID} \tag{39}$$

The set of solution for any $V_2^\top \mathbf{x}$ is given by

$$\widehat{\mathbf{V}^\top \mathbf{x}} = \begin{pmatrix} \mathbf{S}^{-1} U_1^\top \mathcal{Y}_{ID} \\ V_2^\top \mathbf{x} \end{pmatrix} \tag{40}$$

The optimal solution of Eq. (24) is given by :

$$\begin{aligned}
\hat{\mathbf{x}} &= \mathbf{V} \begin{pmatrix} \mathbf{S}^{-1} U_1^\top \mathcal{Y}_{ID} \\ V_2^\top \mathbf{x} \end{pmatrix} \\
&= V_1 \mathbf{S}^{-1} U_1^\top \mathcal{Y}_{ID} + V_2 V_2^\top \mathbf{x}
\end{aligned} \tag{41}$$

6 Control of the eight motors : Feedback controller setting

If the effects of friction are considered negligible during rotation of the output shaft, the mechanical equation of output shaft of each motor is as follows :

$$I \dot{x}_2 = \Gamma \tag{42}$$

The physical parameter I is the moment of inertia of the output shaft for each motor. Its numerical value is equal to $3.1 \cdot 10^{-3} \text{kg.m}^2$. Γ is the torque provided by the output shaft of each motor. For each of the eight motors, a PID corrector is defined to control the angular position of the output shaft such as :

$$\dot{x}_2 = \dot{x}_2^d - K_v(x_2 - x_2^d) - K_p(x_1 - x_1^d) - K_i \int_0^t (x_1 - x_1^d) d\tau \tag{43}$$

where $\bullet^d$ defines the desired variable. With $e = x_1 - x_1^d$ error equation is then :

$$\ddot{e} + K_v \dot{e} + K_p e + K_i \int_0^t e d\tau = 0 \tag{44}$$

With the initial conditions $e(t=0) = e_0$ et $\dot{e}(t=0) = 0$, the introduction of the Laplace operator s leads to the Laplace transform $E(s)$:

$$E(s) = \frac{(s^2 + K_v s) e_0}{s^3 + K_v s^2 + K_p s + K_i} \tag{45}$$

Let us consider that the dynamics of $E(s)$ (45) is governed by a real negative pole $s = -a$ of multiplicity 3 such as :

$$(s + a)^3 = s^3 + 3as^2 + 3a^2s + a^3 \tag{46}$$

Identifying the denominator of (45) with (46) leads to the following choice of PID corrector gains :

$$K_v = 3a, \quad K_p = 3a^2, \quad K_i = a^3.$$

The parameter a was chosen real and taking into account the dynamic limits of each motor, i.e. its electromechanical constant T_{em} . With $a \approx \frac{2}{T_{em}}$, the closed loop system has an attenuation close to -3 dB for $\omega = \frac{1}{T_{em}}$. Its chosen value is equal to 77 rad/s. The applied motor torque is thus :

$$\Gamma = I(\dot{x}_2^d - 3a(x_2 - x_2^d) - 3a^2(x_1 - x_1^d) - a^3 \int_0^t (x_1 - x_1^d) d\tau) \quad (47)$$

Remark : The expression for the torque controlling (47) the motor output shaft is equivalent to computed torque law for a rigid multi-body robot, which is fully actuated [10].

7 Experiments

The platform performs several trajectories that are composed of four orbital movements. A video of one trajectory is available at ¹. A sampling period of 1 ms is used to measure the position vector $\mathbf{t}$ and the cable tension $\boldsymbol{\tau}$ with the *FUTEK FSH04097* sensors. The two objectives of these experiments are firstly to evaluate the performance of the differentiator, which is applied off-line to recorded signals in order to identify the physical parameters of MP, and secondly to test on-line this differentiator for closed-loop control of the eight motors using the PID control law Eq. (43).

7.1 Identification of the physical parameters of the platform

Measurement data are recorded over 52 seconds. The eight measured cable tensions are plotted Fig. 3 a). The Euler angles of the platform during its motion are shown in Fig. 3 b). The variations of these Euler angles are small, of the order of 10^{-4} rad. This motion of the platform is indeed translational as it is predicted by the trajectory planning. The identification of the parameters M , Mx_G , My_G , and Mz_G needs the linear acceleration vector $\ddot{\mathbf{t}}$ of the MP. For the best knowledge of this acceleration vector $\ddot{\mathbf{t}}$ a comparison is made from the measured platform position $\mathbf{t}$ between the estimation of the linear velocity vector $\dot{\mathbf{t}}$ and the linear acceleration vector $\ddot{\mathbf{t}}$ respectively thanks to the proposed SIHD-3 algorithm and the estimation of these vectors $\dot{\mathbf{t}}$ and $\ddot{\mathbf{t}}$ thanks the simplest Euler differentiation method.

The estimation of the velocity vector $\dot{\mathbf{t}}$ and the acceleration vector $\ddot{\mathbf{t}}$ with the SIHD-3 differentiator is summarized Fig. 2. Five parameters are defined for the SIHD-3 algorithm in cascade : λ_1 , λ_2 , and

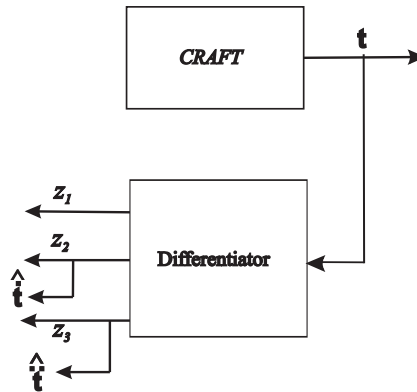


FIGURE 2 – Estimation variables $\hat{\mathbf{t}}$ and $\hat{\ddot{\mathbf{t}}}$ respectively of the linear velocity vector and the linear acceleration vector thanks to the SIHD-3 algorithm

λ_3 for the differentiator, α and μ . The numerical values of these five parameters are set as follows :

$$\lambda_1 = 210, \quad \lambda_2 = 210, \quad \lambda_3 = 525, \quad \alpha = 0.95, \quad \mu = 1 \quad (48)$$

1. <https://youtube/I-IOcAGha3o>

With the simplest Euler differentiation algorithm the numerical tests shows that the estimation of $\dot{\mathbf{t}}$ and $\ddot{\mathbf{t}}$ requires filtering the position vector $\mathbf{t}$ of the platform with a cutoff frequency of 4Hz. The filter applies a zero-phase forward and reverse digital infinite impulse response (*IIR*) filtering. Figure 4 shows the estimated velocity and acceleration vectors in 3D by using the SIHD-3 algorithm and the simplest Euler differentiation algorithm. The curves are similar between the two algorithms. Table 1 gathers the four identified parameters. The standard deviations are not computed based on the standard deviation from the actual parameter values that are unknown. The numerical values obtained by the two methods for each identified parameter are very close to each other. However, having to filter the position measurement of the platform is a strong constraint in the perspective of doing real time identification of the dynamic parameters of *CRAFT*. That is why the SIHD-3 algorithm, which is iterative, is a relevant solution in the perspective of real time identification.

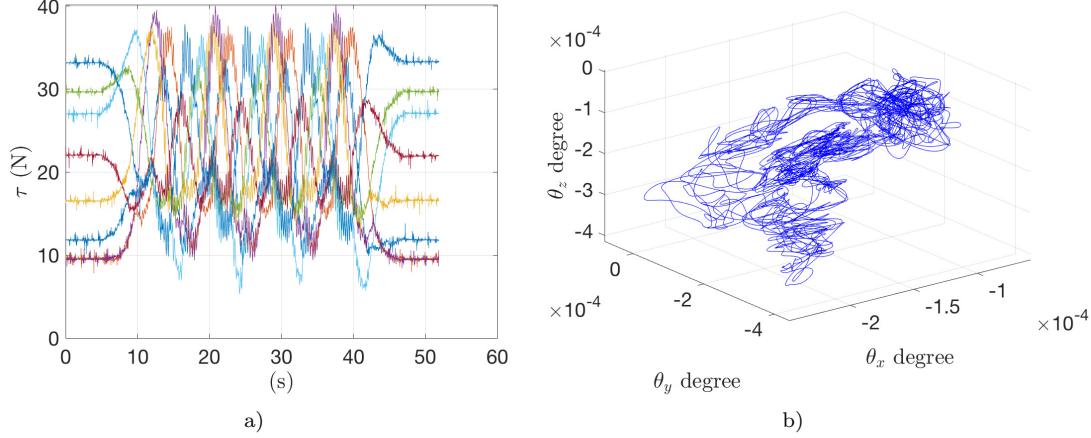


FIGURE 3 – a) Cable tensions τ as function of time, b) Orientation of the mobile platform.

TABLE 1 – Numerical values of the identified parameters

Identified parameters	differentiation SIHD-3			Euler differentiation		
	$\hat{\mathbf{x}}_i$	$\hat{\sigma}_{\mathbf{x}_i}$	$\% \hat{\sigma}_{\mathbf{x}_i}$	$\hat{\mathbf{x}}_i$	$\hat{\sigma}_{\mathbf{x}_i}$	$\% \hat{\sigma}_{\mathbf{x}_i}$
M (kG)	9.88	0.0027	0.027	9.88	0.0027	0.027
Mx_G (kGm)	-0.064	0.0027	4.21	-0.064	0.0027	4.16
My_G (kGm)	0.019	0.0027	13.86	0.019	0.0027	13.65
Mz_G (kGm)	-1.059	0.070	6.67	-1.210	0.078	6.41

7.2 Experiments of different control law versions

The aim here is to compare the efficiency of the differentiator Eq. (21) with that of the finite difference to control the eight motors that operate the cables during the course of the MP trajectory for three different durations, $T_f = 60$ s, $T_f = 40$ s, and $T_f = 20$ s. This comparison is carried out thanks to the three following versions of the PID control law Eq. (43) :

Case 1 :

$$\Gamma = I(\dot{x}_2^d - K_v(\frac{y_m - x_1^-}{h} - x_2^d) - K_p(y_m - x_1^d) - K_i \int_0^t (y_m - x_1^d) d\tau) \quad (49)$$

Case 2 :

$$\Gamma = I(\dot{x}_2^d - K_v(z_2 - x_2^d) - K_p(z_1 - x_1^d) - K_i \int_0^t (z_1 - x_1^d) d\tau) \quad (50)$$

Case 3 :

$$\Gamma = I(\dot{x}_2^d - K_v(z_2 - x_2^d) - K_p(y_m - x_1^d) - K_i \int_0^t (y_m - x_1^d) d\tau) \quad (51)$$

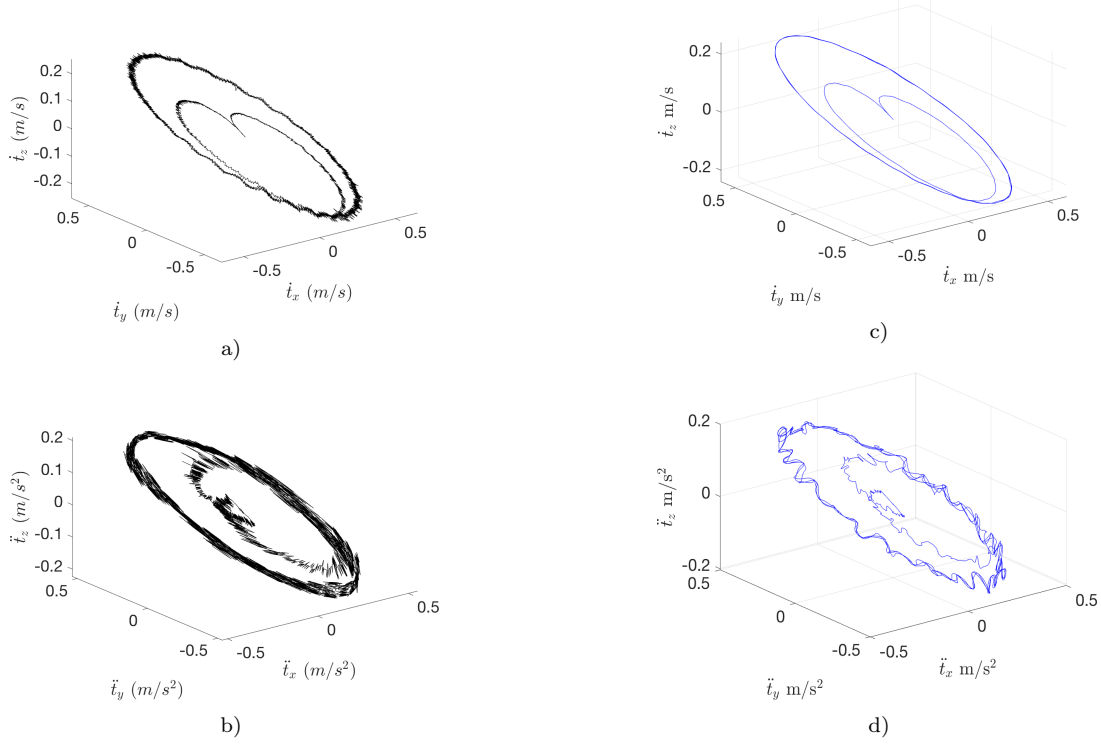


FIGURE 4 – $\hat{\mathbf{t}}$ and $\hat{\dot{\mathbf{t}}}$ along the orbital trajectory of the platform : velocity (a) and acceleration (b) estimated with the SIDH-3 algorithm ; velocity (c) and acceleration (d) computed with the simplest Euler differentiation algorithm.

Remark : The integral terms is calculated with the trapezoidal method such as

$$S = S^- + \frac{(\bullet - x_1^d) + (\bullet - x_1^d)^-}{2} h \quad (52)$$

with $S(t=0) = 0$. In the absence of a tachometer, it is not possible to evaluate the efficiency of the differentiator explicitly. Consequently, the performance of the differentiator was evaluated in terms of the quality of trajectory tracking. The gains of the PID corrector remain unchanged for all tests. This evaluation is based on the following criterion :

$$\mathcal{C} = \frac{1}{T_f} \sum_{j=1}^N e_j^2 h \quad (53)$$

where N is the size of the tracking error vector between the measure and the desired trajectory of the joint variable $\mathbf{e} = \mathbf{y} - \mathbf{x}_1$. In Figure 5, the profiles of this tracking error vector are shown for motor 4 with the three control laws Eq. (49), Eq. (50), and Eq. (51) respectively. The magnitude of these error vectors is small, less than $6.0 \cdot 10^{-3}$ with the three control laws respectively. The order of magnitude of this tracking error vector for the other motors is similar. For the three movements and control laws Eq. (49), Eq. (50), and Eq. (51) the numerical values of the criterion Eq. (53) are gathered in table 2 for each of the eight motors. For a given motor, the criterion values are relatively close. The differentiator therefore has no disturbance effects on the PID control law. For orbital movements 1 and 2, trajectory tracking is globally better with the control law Eq. (51), and even better with the control law Eq. (50) that uses estimates of the joint variable and joint speed than with the control law Eq. (49). When the orbital trajectory is executed more quickly (Orbital movement 3, $T_f=20$ s), its tracking is better with the control law Eq. (49) based on the use of angular variable measurement y_m and finite difference angular velocity calculation.

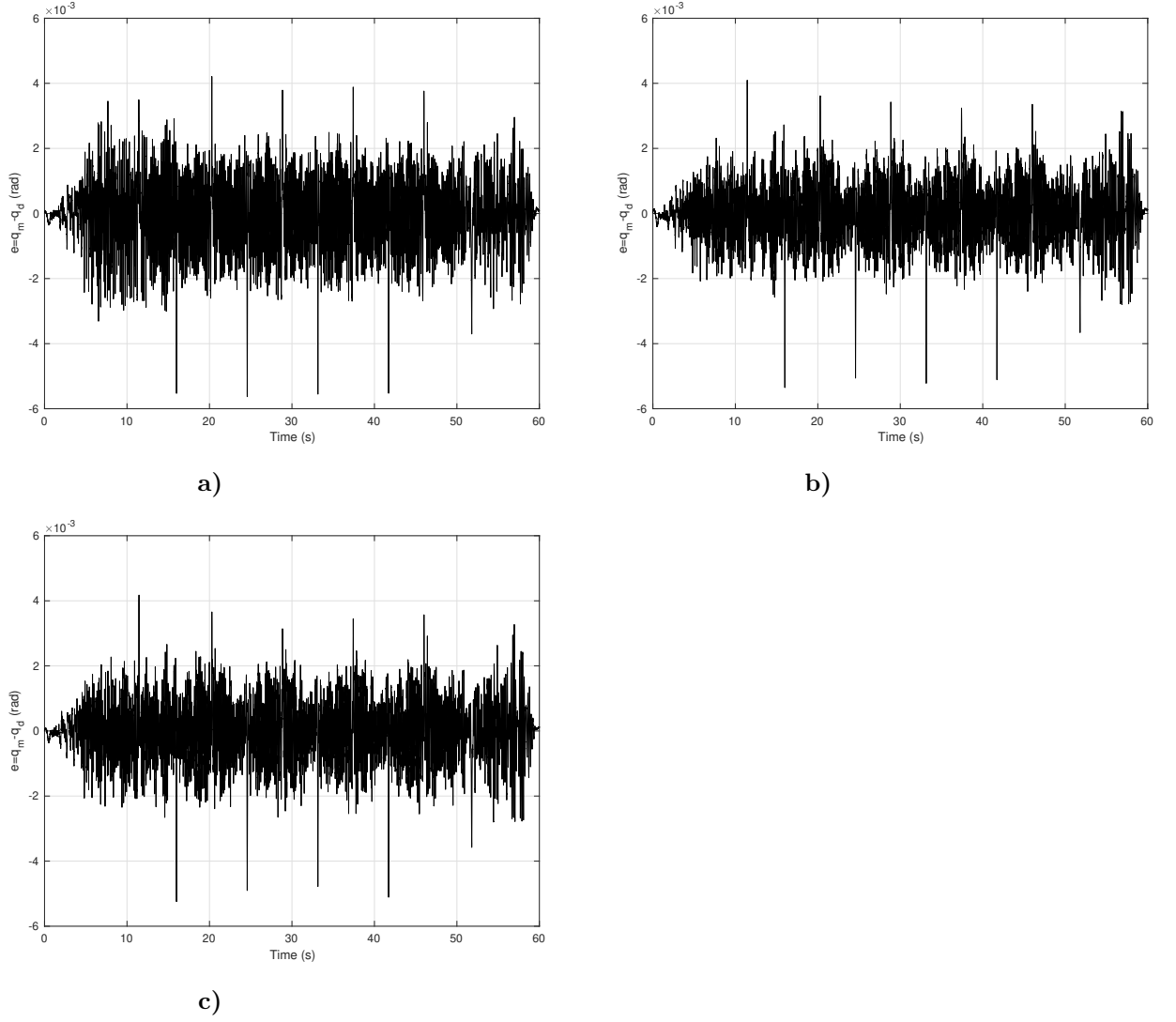


FIGURE 5 – Motor 4 : Evolution of the error tracking for the Orbital movement 1 - $T_f = 60$ s : a) with Eq. (49), b) with Eq. (50), c) with Eq. (51).

TABLE 2 – Trajectory tracking criterion $\mathcal{C}$ Eq. (53) for each motor with the control laws Eq. (49), Eq. (50), and Eq. (51).

Motor	$\mathcal{C}$: Orbital movement 1 ($T_f = 60$ s)			$\mathcal{C}$: Orbital movement 2 ($T_f = 40$ s)			$\mathcal{C}$: Orbital movement 3 ($T_f = 20$ s)		
	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3	Case 1	Case 2	Case 3
1	$6.0 \cdot 10^{-7}$	$5.6 \cdot 10^{-7}$	$7.3 \cdot 10^{-7}$	$8.0 \cdot 10^{-7}$	$1.00 \cdot 10^{-6}$	$1.33 \cdot 10^{-6}$	$4.23 \cdot 10^{-6}$	$4.52 \cdot 10^{-6}$	$5.00 \cdot 10^{-6}$
2	$8.9 \cdot 10^{-7}$	$7.9 \cdot 10^{-7}$	$7.9 \cdot 10^{-7}$	$9.0 \cdot 10^{-7}$	$1.00 \cdot 10^{-6}$	$1.00 \cdot 10^{-6}$	$3.27 \cdot 10^{-6}$	$5.39 \cdot 10^{-6}$	$5.59 \cdot 10^{-6}$
3	$8.6 \cdot 10^{-7}$	$9.4 \cdot 10^{-7}$	$7.9 \cdot 10^{-7}$	$1.20 \cdot 10^{-6}$	$1.13 \cdot 10^{-6}$	$1.20 \cdot 10^{-6}$	$3.44 \cdot 10^{-6}$	$3.69 \cdot 10^{-6}$	$3.49 \cdot 10^{-6}$
4	$1.12 \cdot 10^{-6}$	$8.6 \cdot 10^{-7}$	$9.4 \cdot 10^{-7}$	$1.26 \cdot 10^{-6}$	$1.15 \cdot 10^{-6}$	$1.33 \cdot 10^{-6}$	$4.29 \cdot 10^{-6}$	$4.62 \cdot 10^{-6}$	$34.40 \cdot 10^{-6}$
5	$9.9 \cdot 10^{-7}$	$8.2 \cdot 10^{-7}$	$9.4 \cdot 10^{-7}$	$1.08 \cdot 10^{-6}$	$1.14 \cdot 10^{-6}$	$1.16 \cdot 10^{-6}$	$2.99 \cdot 10^{-6}$	$3.19 \cdot 10^{-6}$	$3.12 \cdot 10^{-6}$
6	$1.15 \cdot 10^{-6}$	$1.10 \cdot 10^{-6}$	$9.9 \cdot 10^{-7}$	$1.35 \cdot 10^{-6}$	$1.42 \cdot 10^{-6}$	$1.45 \cdot 10^{-6}$	$3.24 \cdot 10^{-6}$	$3.63 \cdot 10^{-6}$	$3.54 \cdot 10^{-6}$
7	$9.1 \cdot 10^{-7}$	$6.6 \cdot 10^{-7}$	$7.9 \cdot 10^{-7}$	$1.00 \cdot 10^{-6}$	$8.4 \cdot 10^{-7}$	$9.5 \cdot 10^{-7}$	$4.45 \cdot 10^{-6}$	$4.60 \cdot 10^{-6}$	$4.95 \cdot 10^{-6}$
8	$9.5 \cdot 10^{-7}$	$9.1 \cdot 10^{-7}$	$8.4 \cdot 10^{-7}$	$1.06 \cdot 10^{-6}$	$1.28 \cdot 10^{-6}$	$1.24 \cdot 10^{-6}$	$6.70 \cdot 10^{-6}$	$7.11 \cdot 10^{-6}$	$7.37 \cdot 10^{-6}$

8 Discussion and Future Work

This paper presents the definition and use of a SHID-3 differentiator for two tasks, the identification of four physical parameters of the *CRAFT* and the control of its motors. The results are compared with those obtained with the usual finite difference method. For the identification task, the focus is on four parameters, the platform mass and the three components of the first moment of inertia around the origin of the reference frame of the platform, using an orbital trajectory that has been specifically designed to achieve pure platform translation movement in all the three xyz -plane. The trajectory, which occurs within a single plane, allows for the identification of these four dynamic parameters. The values determined by the identification of the inertial parameters are consistent with the actual platform mass (measured) and the estimated center of mass location from the computer aided Design. The results have also been compared with other types of trajectories, such as point-to-point linear trajectories. The reason for choosing a circular trajectory is the availability of continuous platform accelerations which facilitates the use of the proposed methodology. The results obtained from experimental measurements of the platform position, as well as measurements treated with the application of a homogeneous semi-implicit differentiator, indicates that despite the complexity of the *CRAFT*. These results allow the real time identification of all the essential dynamic parameters.

The second task is the feedback control of the eight motors and to evaluate how this SHID-3 differentiator enhances the control to track a circular trajectory. Even if it is not possible to claim a net positive effect of using the SHID-3 differentiator for closed-loop motor control compared with finite-difference angular-velocity calculation, the results are quite respectable.

In future work, the proposed approach will be used to identify the remaining six dynamic terms associated with the inertia tensor, specifically three terms associated with the platform moment of inertia and other three are products of inertia. To manage the future identification tasks, an IMU sensor will be used in order to provide acceleration measurements that are synchronized with those of the cable tensions. Additionally, we will also aim to identify the friction that acts throughout the actuators of *CRAFT*. For the control, in-depth work is needed to define the gains of both the feedback control and the differentiator. This will be all the more true when we integrate the MP dynamics into a global control scheme to position it in space.

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